Experiment No. 7

SOLAR CELL CHARACTERISTICS

Aim: To study the characteristics of solar cell and calculate the Fill Factor.

Apparatus: Board mounted with solar cell, bulb, DC power supply, digital multimeter, ammeter (0-50mA.)

Formula:

1.Fill Factor, F = $\boxed{\dfrac{\text{Workable wattage}}{\text{Ideal wattage}} = \dfrac{V_w I_w}{V_{oc} I_{sc}}}$

Where, V = Workable voltage (volt)

I = Workable current (mA)

V_{oc}= Open circuit voltage (volt)

I_{sc} =short circuit current (mA)

2.Series resistance (ohm)

$R_s = Slop\ of\ V\ vs\ \Delta I$

3.Shunt resistance (ohm)

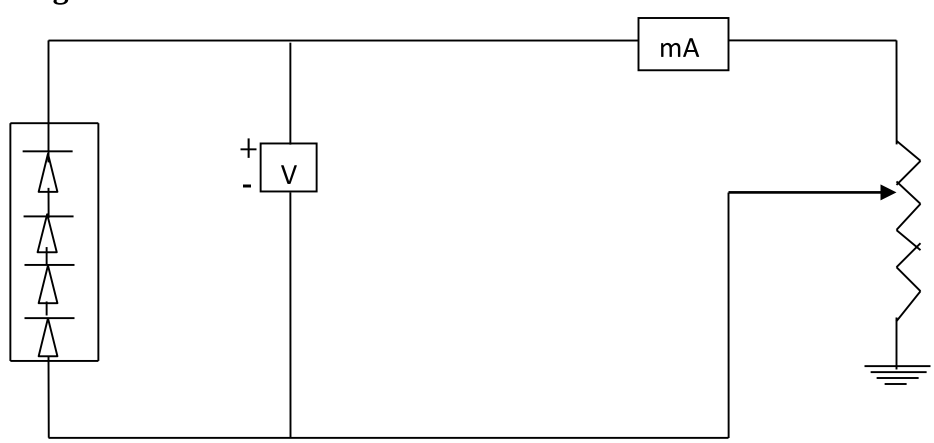
$R_p = Slopeof\ \Delta V vs\ I\ curve$

Procedure: -

1. Connect the circuit as shown in the figure.
2. Switch on the apparatus with the lamp casting light on the cell.
3. Use multi-meters as voltmeter and ammeter for measuring voltage and current respectively.

4. Measure I_{sc} for 0 load (R_L= 0).
5. Vary R_L and note corresponding values of voltage and current.
6. Measure V_{oc} (Voltage when R_L is maximum).
7. Plot I-V curve. Taking V on x-axis and I on y-axis.
8. Calculate power output. Power is V*I.
9. Plot power (P) Vs potential (V) in the same I-V curve taking V on x-axis.
10. Consider the point where power is maximum. The corresponding voltage is V_w, draw a line parallel to I-axis, at maximum power, where it cut the I-V curve that current will be I_w.
11. Calculate the fill factor (F).
12. Find V vs ΔI values and ΔV vs I values and complete Observation Table 2 and 3
13. Plot a graph of V vs ΔI, find slope of the graph and calculate R_S= slope.
14. Also calculate R_P from the graph ΔV vs I

Circuit diagram: -



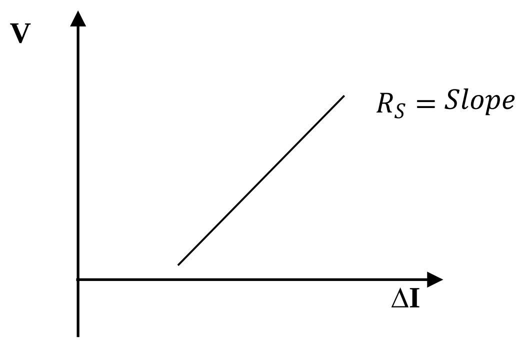
Observation Table: -

Observation Table 1 for obtaining Fill Factor

Serial No.	Resistance (Rs) Ω	Voltage across solar cell (V)	Current in the circuit (mA)
1			
2			
.			
.			
.			
20			

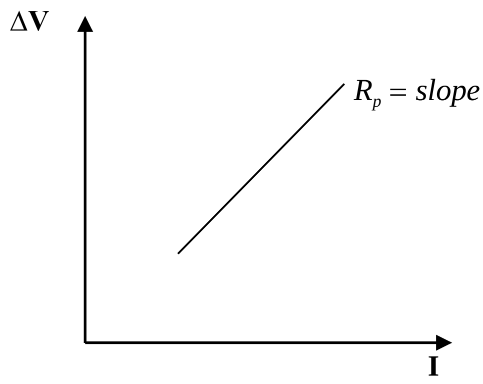
Observation Table 2 for obtaining Series Resistance R_s

Sr.No	Voltage (V)	ΔI



Observation Table 3 for obtaining Shunt Resistance R_p

Sr.No	ΔV (volt)	I(mA)



Calculation:

Result:

Fill factor of given solar cell =

Shunt resistance, R_P = ohm

Series resistance, R_S = ohm

Precaution:

1. The solar cell should be exposed to sun light before using it in the experiment.

2. Light from the lamp should fall normally on the cell

3. A resistance in the cell circuit should be introduced so that the current does not exceed the safe operating limit.